Addition of 2 odd numbers give even numbers.

Ex: 3+5=8

5+9=22

The examples cannot be substitute of your proof.

Set.

Index set ↓

$\Lambda = \{S, M, T, W, Th, Fri, Sat\}$

$L_{sun} = \{Rice, Dal, Curry\}$

$L_{mon} = \{Bread, Fruit\}$

$L_{sat} = \{\ \}$

$L_{sun} \cup L_{mon} \cup L_{sat}$ ← Instead of writing this you can write.

$\bigcup_{\alpha \in \Lambda} L_\alpha$

Examples:-

$\Lambda = \{1, 2, 3, \dots\}$

$A_7 = [0, 7]$ and $7 \in \Lambda$

Find $\bigcup_{\alpha \in \Lambda} A_\alpha$

a) ϕ Ans:- $[0, \alpha]$

b) $\{0, 1, 2, 3, \dots\}$

c) $[0, 10000]$

d) None of the above ✓

Find $\bigcap_{\alpha \in \Lambda} A_\alpha$ - $[0, 1]$ →

$A_J = \left[0, \frac{7}{J-1}\right]$ - Find at home.

7 7 7

Write few sets:

$\alpha = 1, 2, 3, \dots$

when $\alpha = 1$, $A_1 = [0, 1]$

when $\alpha = 2$, $A_2 = [0, 2)$

∴ we need to find.

$[0,1] \cap [0,2) \cap [0,3) \dots$

The smallest interval is $[0,1]$

Every other interval contains all the numbers from 0 to 1.

∴ The numbers common to all the intervals are exactly the numbers in $[0,1]$

Hence, $\bigcap_{\alpha \in A} A_\alpha = [0,1]$

$\Lambda = \{x | x \text{ is the set of } \mathbb{R}\}$

$A\lambda = \{1, -2, 2\}$

$\bigcup_{\alpha \in A} A_\alpha, \bigcap_{\alpha \in A} A_\alpha$

$A\alpha$ means the set corresponding to index Λ.

$(A_1 \cap A_2)^c = A_1^c \cup A_2^c$ $\quad \Lambda = \{1, 2\}$

$\Lambda = \{1, 2, 3, \ldots\}$ - Index set

$(\bigcap_{\alpha \in \Lambda} A_\alpha)^c = \bigcup_{\alpha \in \Lambda} A_\alpha^c$ → Prove

$y \in (\bigcap_{\alpha \in \Lambda} A_\alpha)^c$ $\quad$ (Λ - capital lamda.)

$\Rightarrow y \notin \bigcap_{\alpha \in \Lambda} A_\alpha$ (Properties of complement)

$\Rightarrow \exists$ atleast $\alpha \in \Lambda, \Lambda \mid y \notin \Lambda\alpha$. (properties of intersection)

$\Rightarrow y \in A_\alpha^c$

$\rightarrow y \in \bigcup_{\alpha \in \Lambda} A_\alpha^c$

Cardinality of the set - Number of unique elements

$A = \{1, 2, 3\}$

$A = \{1, 1, 2, 2, 3, 3\}$

} They are identical.

DeMorgans Law:

$$\left(\bigcap_{\alpha \in \Lambda} A_\alpha\right)^c = \bigcup_{\alpha \in \Lambda} A_\alpha^c$$

$$\left(\bigcup_{\alpha \in \Lambda} A_\alpha\right)^c = \bigcap_{\alpha \in A} A_\alpha^c$$

Complements of intersection → union of complements

$$(\cap A)^c = \cup A^c$$

Complement of union → intersection of complements

$$(\cup A)^c = \cap A^c$$

Homework.

let $A = \{0,1,2,3,\cdots\}$

For $I = 0,1,2\cdots$

a) let $A_I = [0, \frac{I}{I-1}]$

b) let $A_I = \{-I, I\}$

In Both cases, find $\cup_{I \in A} I$ and $\cap_{I \in A} I$

a) For $I = 0$; $A_0 = [0, \frac{0}{0+1}] = [0,0] = \{0\}$

For $I = 1$; $A_1 = [0, \frac{1}{1+1}] = [0, \frac{1}{2}]$

$A_0 \subseteq A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots$

$\cup_{I \in A} I = [0,1]$

For any $I \geq 0$, $0 \leq \frac{I}{I+1}$, so, $0 \in A_I$ for all I

$\cap_{I \in A} I = \{0\}$

b) $A_I = \{-I, I\}$

For $I = 0$, $A_0 = \{-0,0\} = \{0\}$

For $I = 1$, $A_1 = \{-1,1\}$

For $I = 2$, $A_2 = \{-2,2\}$

$\cup_{I \in A} I = A_0 \cup A_1 \cup A_2 \cdots$

$= \{0,1,-1,2,-2\cdots\}$ - contains set of all integers

$\therefore \cup_{I \in A} I = \mathbb{Z}$ (set of all integers)

$\cap_{I \in A} I = \emptyset$ (No single element exists in all A sets across $I = 0,1,2\cdots$ the intersection is empty)